

Data Visualization Dashboard on Mobile Sales Dataset

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1 Introduction

Data visualization is an important part of data analysis and business intelligence. It helps transform raw data into clear charts, dashboards, and insights that support better decision-making.

This project focuses on creating a sales analytics dashboard using a Mobile and Laptop Sales Dataset. The dataset contains product, brand, price, quantity sold, region, and date-related information. The main goal of this task is to visualize business performance and identify meaningful sales trends and patterns.

2 Objective

The objective of this project is to create visualizations that present business insights from the dataset.

The main goals are:

- Load and explore the sales dataset using Pandas.
- Perform basic data cleaning and preprocessing.
- Create KPI cards for important business metrics.
- Build bar charts, pie charts, histograms, line charts, and correlation heatmaps.
- Analyze sales performance by brand, product, region, and time.
- Create a complete dashboard for business insight presentation.

3 Dataset Description

The dataset contains sales records related to mobile phones and laptops. It includes approximately 50,000 records and several important columns.

The main dataset attributes include:

- Product
- Brand

- Product Code
- Product Specification
- Price
- Inward Date
- Dispatch Date
- Quantity Sold
- Customer Name
- Customer Location
- Region
- Core Specification
- Processor Specification
- RAM
- ROM
- SSD

This dataset is suitable for sales analytics, product performance analysis, regional analysis, and dashboard development.

4 Tools and Technologies Used

The following tools and technologies were used in this project:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

5 Methodology

The project was completed through several steps.

5.1 Data Loading

The dataset was loaded using Pandas. After loading the dataset, the first rows, dataset shape, column names, and data types were inspected.

5.2 Data Cleaning

Missing values were checked using Pandas. Some columns contained a very small number of missing values, while specification-related columns had missing values because not all specifications apply to every product type.

The following cleaning steps were performed:

- Checked missing values.
- Removed rows with critical missing values.
- Filled non-applicable specification columns with appropriate values.
- Converted date columns into datetime format.
- Created a new Revenue column.

The Revenue column was calculated as:

[Revenue = Price $\times$ *Quantity Sold*]

5.3 KPI Calculation

KPI cards were created to summarize the most important business metrics.

The main KPIs included:

- Total Revenue
- Total Units Sold
- Average Product Price
- Top Selling Brand

6 KPI Cards

KPI cards provide a quick overview of overall sales performance. They help users understand the most important business numbers without looking through the entire dataset.



Figure 1: KPI Cards for Sales Dashboard

7 Dashboard Visualizations

Several visualizations were created to analyze the dataset from different business perspectives.

7.1 Top 10 Brands by Revenue

A bar chart was created to identify the top-performing brands based on total revenue. This visualization helps understand which brands contribute the most to overall sales.

7.2 Product Distribution

A pie chart was used to show the distribution of product categories in the dataset. This helps compare the share of mobile phones and laptops.

7.3 Price Distribution

A histogram was created to analyze the distribution of product prices. This visualization helps understand the general pricing structure of the products.

7.4 Revenue by Region

A bar chart was used to compare revenue across different regions. This provides insight into which regions generate higher sales.

7.5 Monthly Revenue Trend

A line chart was created to show how revenue changes over time. This helps identify sales trends and monthly performance patterns.

7.6 Correlation Heatmap

A correlation heatmap was created to show relationships between numerical variables such as price, quantity sold, and revenue.

8 Final Sales Dashboard

The final dashboard combines multiple visualizations into one complete view. It provides a clear summary of brand performance, product distribution, price patterns, regional revenue, monthly sales trends, and variable correlations.

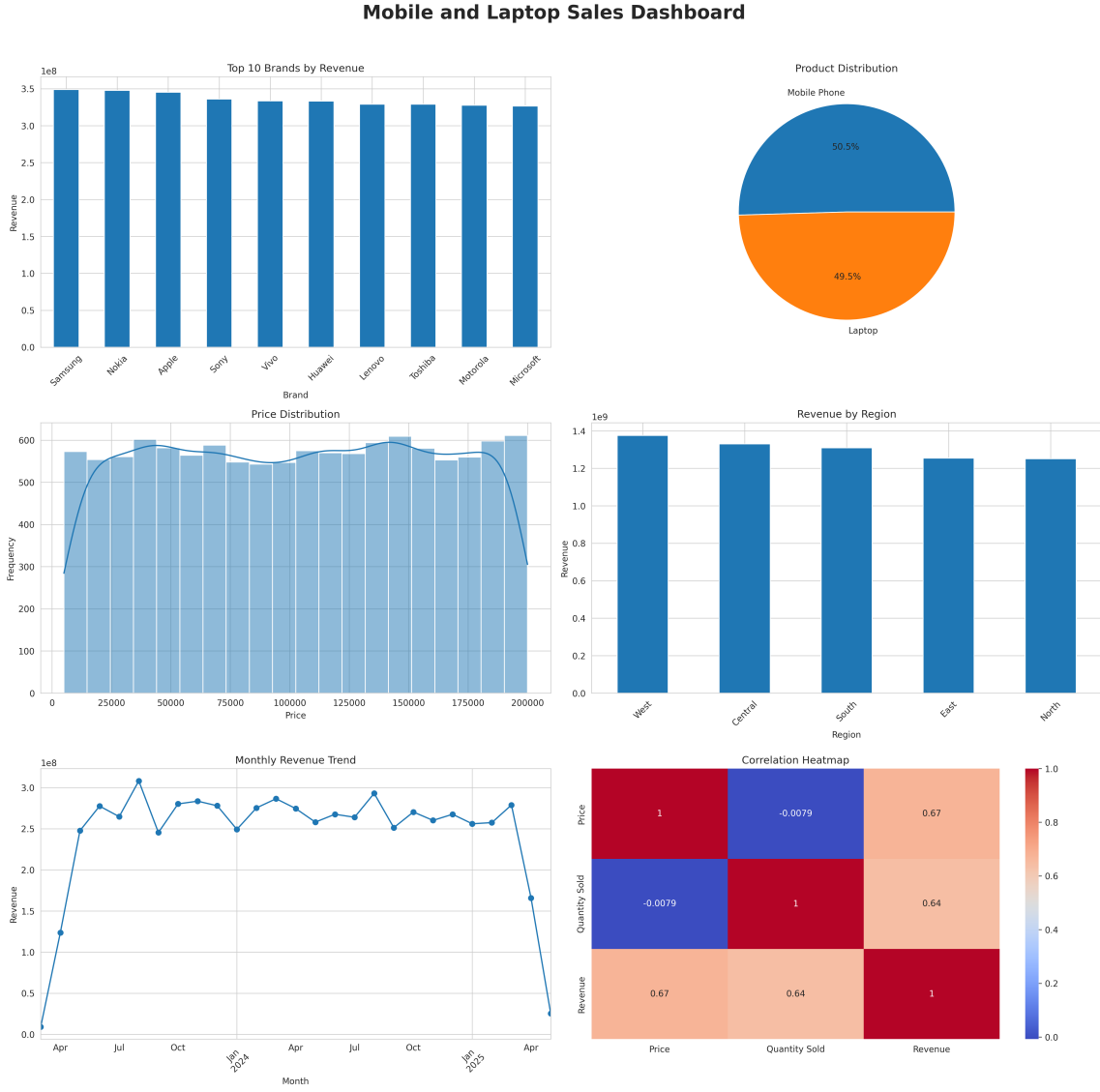


Figure 2: Final Mobile and Laptop Sales Dashboard

9 Business Insights

The dashboard helped identify several important business insights:

1. Total revenue and total units sold provide a clear overview of overall sales performance.
2. Top brand analysis helps identify which brands generate the highest revenue.
3. Product distribution shows the share of different product types in the dataset.
4. Regional revenue analysis helps identify high-performing sales regions.
5. Monthly revenue trend shows how sales performance changes over time.
6. Price distribution helps understand the pricing structure of products.

7. Correlation analysis shows relationships between numerical variables such as price, quantity sold, and revenue.

10 Conclusion

This project successfully created a complete Data Visualization Dashboard using Python. The analysis transformed raw sales data into meaningful business insights through KPI cards, charts, and a final dashboard.

The project helped strengthen practical skills in data visualization, dashboard design, business intelligence, and trend analysis. It also demonstrated how visual analytics can support better decision-making in sales and product performance analysis.

11 GitHub Repository

The full project, source code, dataset, dashboard images, and report are available on GitHub:

<https://github.com/Nazzekax/mobile-sales-dashboard>